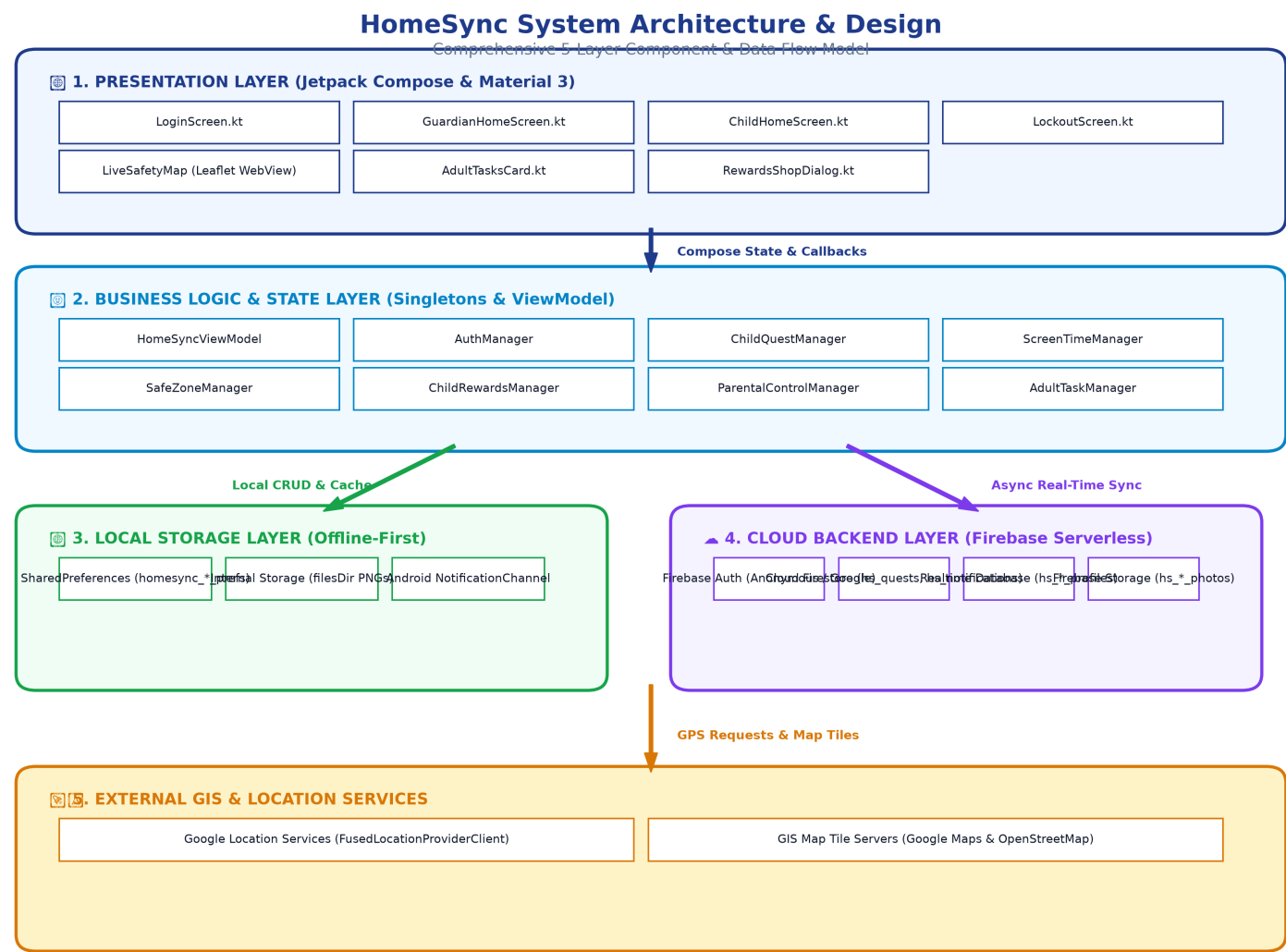


HomeSync System Architecture & Design

High-Level 5-Layer System Architecture & Data Flow Diagram



Layer Name	Key Responsibilities & Components
1. Presentation Layer	Renders Jetpack Compose screens (Login, Guardian, Child, Lockout, Arcade, Leaflet WebView Map).
2. Business Logic Layer	Manages state (HomeSyncViewModel) and business singletons (Auth, Quests, ScreenTime, SafeZones).
3. Local Storage Layer	Offline-first SharedPreferences (homesync_*_prefs), internal storage files, and NotificationChannel.

4. Cloud Backend Layer	Firestore Auth, Cloud Firestore (hs_quests), Realtime DB (hs_*_profiles), and Cloud Storage.
5. External GIS Services	Google FusedLocationProviderClient GPS tracking and OpenStreetMap / Google Maps tile mirrors.